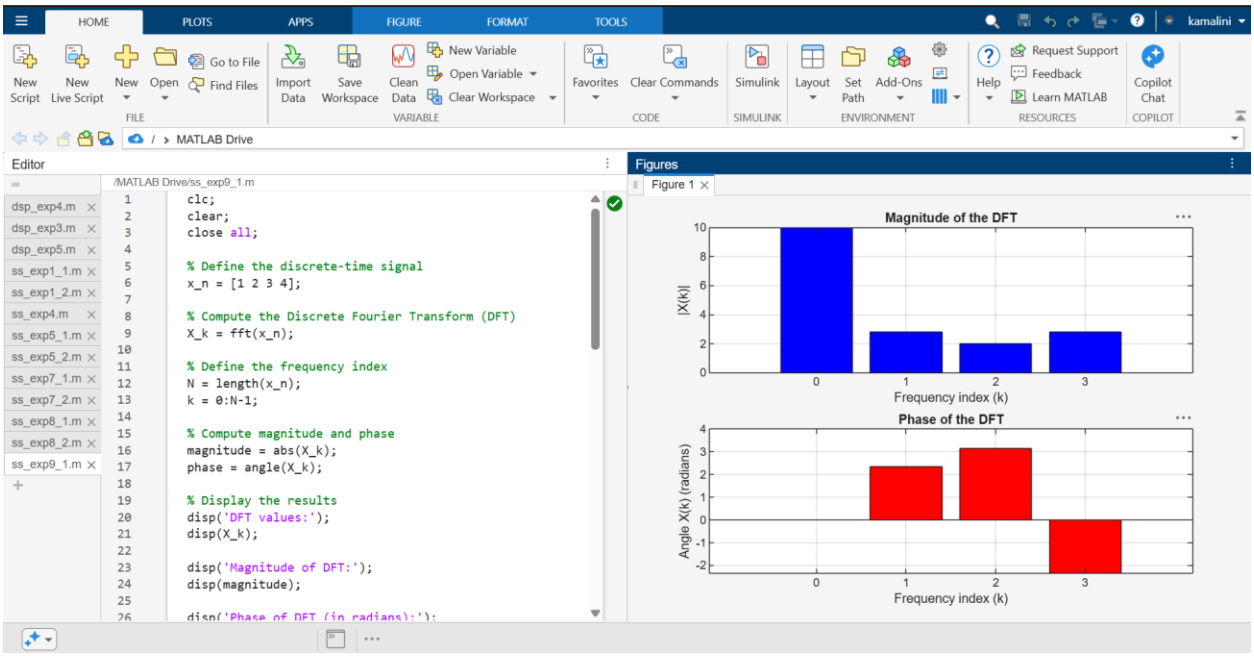
EXPERIMENT 9

```
clc;
clear;
close all;
% Define the discrete-time signal
x_n = [1 2 3 4];
% Compute the Discrete Fourier Transform (DFT)
X_k = fft(x_n);
% Define the frequency index
N = length(x_n);
k = 0:N-1;
% Compute magnitude and phase
magnitude = abs(X_k);
phase = angle(X_k);
% Display the results
disp('DFT values:');
disp(X_k);
disp('Magnitude of DFT:');
disp(magnitude);
disp('Phase of DFT (in radians):');
disp(phase);
% Plot magnitude and phase
figure;
% Magnitude plot
subplot(2,1,1);
bar(k, magnitude, 'b');
xlabel('Frequency index (k)');
ylabel('|X(k)|');
title('Magnitude of the DFT');
grid on;
% Phase plot
subplot(2,1,2);
bar(k, phase, 'r');
xlabel('Frequency index (k)');
ylabel('Angle X(k) (radians)');
title('Phase of the DFT');
grid on;
```


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
kamalini
New Script
New Live Script
New
Open
Go to File
Find Files
FILE
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
VARIABLE
Favorites
Clear Commands
CODE
Simulink
SIMULINK
Layout
Set Path
Add-Ons
ENVIRONMENT
Help
Request Support
Feedback
Learn MATLAB
RESOURCES
Copilot Chat
COPILOT
/ › MATLAB Drive
Editor
/MATLAB Drive/ss_exp9_1.m
dsp_exp4.m
dsp_exp3.m
dsp_exp5.m
ss_exp1_1.m
ss_exp1_2.m
ss_exp4.m
ss_exp5_1.m
ss_exp5_2.m
ss_exp7_1.m
ss_exp7_2.m
ss_exp8_1.m
ss_exp8_2.m
ss_exp9_1.m
1 clc;
2 clear;
3 close all;
4
5 % Define the discrete-time signal
6 x_n = [1 2 3 4];
7
8 % Compute the Discrete Fourier Transform (DFT)
9 X_k = fft(x_n);
10
11 % Define the frequency index
12 N = length(x_n);
13 k = 0:N-1;
14
15 % Compute magnitude and phase
16 magnitude = abs(X_k);
17 phase = angle(X_k);
18
19 % Display the results
20 disp('DFT values:');
21 disp(X_k);
22
23 disp('Magnitude of DFT:');
24 disp(magnitude);
25
Figures
Figure 1
Magnitude of the DFT
|X(k)|
Frequency index (k)
Phase of the DFT
Angle X(k) (radians)
Frequency index (k)